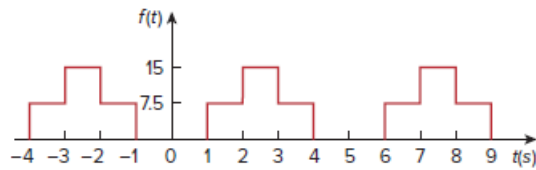


Problem

Find the Fourier series representation of the signal shown in Fig. 17.64.

Figure 17.64:



Step-by-step solution

Step 1 of 4

Refer to Equation 17.3 in the textbook.

The Fourier series is,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Where,

$$\omega_0 = \frac{2\pi}{T} \text{ is the fundamental frequency.}$$

The function is,

$$f(t) = \begin{cases} 0, & 0 < t < 1 \\ 7.5, & 1 < t < 2 \\ 15, & 2 < t < 3 \\ 7.5, & 3 < t < 4 \\ 0, & 4 < t < 5 \end{cases}$$

Step 2 of 4

The time period is, $T = 5$.

Determine the value of ω_0 .

$$\omega_0 = \frac{2\pi}{5}$$

Determine the value of a_0 .

$$\begin{aligned} a_0 &= \frac{1}{T} \int_0^T f(t) dt \\ &= \frac{1}{5} \int_0^5 f(t) dt \\ &= \frac{1}{5} \int_0^1 f(t) dt + \frac{1}{5} \int_1^2 f(t) dt + \frac{1}{5} \int_2^3 f(t) dt + \frac{1}{5} \int_3^4 f(t) dt + \frac{1}{5} \int_4^5 f(t) dt \\ &= \frac{1}{5} \int_0^1 0 \cdot dt + \frac{1}{5} \int_1^2 7.5 dt + \frac{1}{5} \int_2^3 15 dt + \frac{1}{5} \int_3^4 7.5 dt + \frac{1}{5} \int_4^5 0 \cdot dt \\ &= 1.5 + 3 + 1.5 \\ &= 6 \end{aligned}$$

Determine the value of a_n .

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt \\
 &= \frac{2}{5} \int_0^5 f(t) \cos(n\omega_0 t) dt \\
 &= \left[\frac{2}{5} \int_0^1 f(t) \cos(n\omega_0 t) dt + \frac{2}{5} \int_1^2 f(t) \cos(n\omega_0 t) dt + \frac{2}{5} \int_2^3 f(t) \cos(n\omega_0 t) dt \right. \\
 &\quad \left. + \frac{2}{5} \int_3^4 f(t) \cos(n\omega_0 t) dt + \frac{2}{5} \int_4^5 f(t) \cos(n\omega_0 t) dt \right] \\
 &= \left[\frac{2}{5} \int_0^1 0 \cdot \cos\left(\frac{2n\pi}{5} t\right) dt + \frac{2}{5} \int_1^2 7.5 \cos\left(\frac{2n\pi}{5} t\right) dt + \frac{2}{5} \int_2^3 15 \cos\left(\frac{2n\pi}{5} t\right) dt \right. \\
 &\quad \left. + \frac{2}{5} \int_3^4 7.5 \cos\left(\frac{2n\pi}{5} t\right) dt + \frac{2}{5} \int_4^5 0 \cdot \cos\left(\frac{2n\pi}{5} t\right) dt \right] \\
 &= 3 \left(\frac{5}{2n\pi} \right) \left[\sin\left(\frac{2n\pi}{5} t\right) \right]_{t=1}^{t=2} + 6 \left(\frac{5}{2n\pi} \right) \left[\sin\left(\frac{2n\pi}{5} t\right) \right]_{t=2}^{t=3} + 3 \left(\frac{5}{2n\pi} \right) \left[\sin\left(\frac{2n\pi}{5} t\right) \right]_{t=3}^{t=4} \\
 &= \left[\frac{15}{2n\pi} \left[\sin\left(\frac{4n\pi}{5}\right) - \sin\left(\frac{2n\pi}{5}\right) \right] + \frac{30}{2n\pi} \left[\sin\left(\frac{6n\pi}{5}\right) - \sin\left(\frac{4n\pi}{5}\right) \right] \right. \\
 &\quad \left. + \frac{15}{2n\pi} \left[\sin\left(\frac{8n\pi}{5}\right) - \sin\left(\frac{6n\pi}{5}\right) \right] \right] \\
 &= \frac{15}{2n\pi} \sin\left(\frac{6n\pi}{5}\right) - \frac{15}{2n\pi} \sin\left(\frac{4n\pi}{5}\right) - \frac{15}{2n\pi} \sin\left(\frac{2n\pi}{5}\right) + \frac{15}{2n\pi} \sin\left(2n\pi - \frac{2n\pi}{5}\right) \\
 &= \frac{15}{2n\pi} \sin\left(\frac{6n\pi}{5}\right) - \frac{15}{2n\pi} \sin\left(\frac{4n\pi}{5}\right) - \frac{15}{n\pi} \sin\left(\frac{2n\pi}{5}\right)
 \end{aligned}$$

Since the function is even, $b_n = 0$.

The Fourier series is,

$$\begin{aligned}
 f(t) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\
 &= 6 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t) \\
 &= 6 + \sum_{n=1}^{\infty} \left\{ \left[\frac{15}{2n\pi} \sin\left(\frac{6n\pi}{5}\right) - \frac{15}{2n\pi} \sin\left(\frac{4n\pi}{5}\right) - \frac{15}{n\pi} \sin\left(\frac{2n\pi}{5}\right) \right] \cos\left(\frac{2n\pi}{5} t\right) \right\}
 \end{aligned}$$

Thus, the Fourier series of the periodic function is,

$$f(t) = 6 + \sum_{n=1}^{\infty} \left\{ \left[\frac{15}{2n\pi} \sin\left(\frac{6n\pi}{5}\right) - \frac{15}{2n\pi} \sin\left(\frac{4n\pi}{5}\right) - \frac{15}{n\pi} \sin\left(\frac{2n\pi}{5}\right) \right] \cos\left(\frac{2n\pi}{5} t\right) \right\}.$$